Power Series and Taylor Series

ANSWER KEY

Radius and interval of convergence

1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.
Ratio test: $\left|\frac{x^{n+1}/(n+1)}{x^n/n}\right| \to |x|$. Converges for $|x| < 1$: radius of convergence $R = 1$.

2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
Ratio test: $\left|\frac{x^{n+1}n!}{(n+1)!\,x^n}\right| = \frac{|x|}{n+1} \to 0$ for every x: radius of convergence $R = \infty$.

Taylor and Maclaurin series

3 Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.
$e^x = 1 + x + \frac{x^2}{2!} + \cdots = 1 + x + \frac{x^2}{2} + \cdots$.

4 Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.
$f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$: $T_3(x) = 1 - \frac{x^2}{2}$.

5 Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.
$0.2 - \frac{(0.2)^3}{6} = 0.2 - 0.001333\ldots \approx 0.1987$.